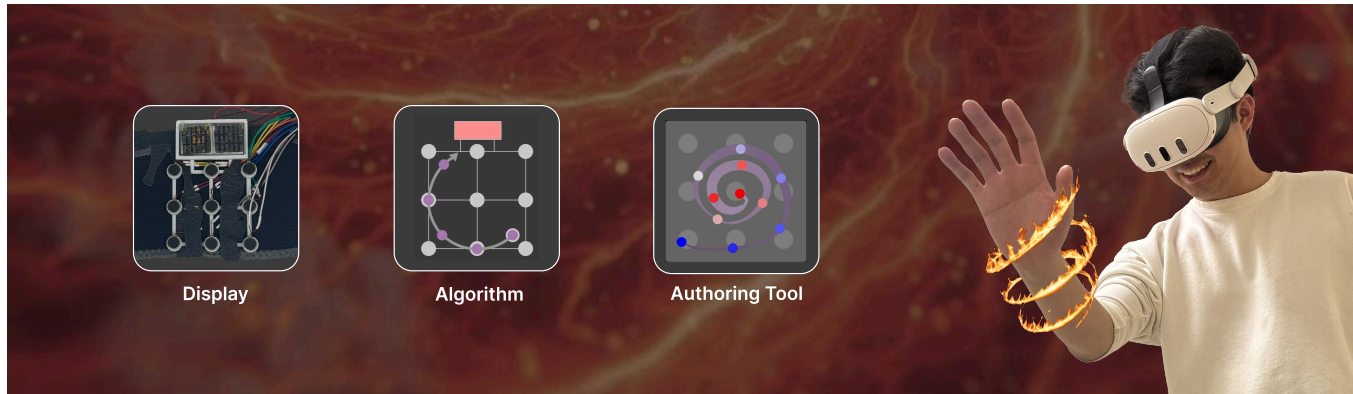


# Demonstration of HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements

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**Figure 1: HeatFlow is a multimodal display, algorithm, and authoring tool that renders dynamic thermal movements by combining thermal and tactile feedback, enriching immersive experiences with perceptual storytelling.**

## Abstract

We present a novel thermal display that integrates thermal and tactile sensations to render dynamic thermal movements across a 2D surface. Leveraging a perceptually validated approach, HeatFlow combines vibration-induced thermal referral and apparent tactile motion to induce the perception of illusory thermal flows. Our demonstration employs a wrappable sleeve form factor to deliver effects to the forearm, palm, and face. 2D flow generation is supported by a novel thermal-tactile control algorithm. In addition to showcasing 2D thermal movements, our demo includes an authoring tool connected to virtual reality scenarios, enabling participants to design custom thermal flows for immersive experiences. Participants will wear the HeatFlow display, iteratively design unique flow patterns with the authoring tool, and directly integrate their flow into a VR scene. This end-to-end design experience highlights the versatility and potential of HeatFlow as a tool for rich multimodal rendering and affective storytelling.

## CCS Concepts

• **Human-centered computing** → **Interaction techniques; Haptic devices; Virtual reality.**

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## Keywords

Thermal-tactile Interface, 2D Thermal Motion, 2D Tactile Motion, Thermal Referral

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## 1 Introduction

Thermal feedback is an exciting frontier for sensory stimulation in immersive experiences. Warm sunlight, chilly wind, burning fire – everyday life is rich with sensory experiences informed by thermal sensations, specifically sensations that move. Movement provides fundamental context for sensory experiences beyond static interaction, especially across spatial trajectories and dynamic changes over time. However, challenges remain in rendering moving thermal feedback with the same level of nuance and depth as in real life, stemming from limitations in current interface design. By overcoming these challenges and expanding the sensory capabilities of rendering devices, opportunities to enhance presence and realism in virtual reality (VR) emerge, promising to transform digital interactions into lifelike and affective experiences.

A naive approach towards thermal motion involves dense arrays of thermal actuators, physically rendering temperature across space similar to existing tactile displays. However, this leads to issues in scalability and comfort, as thermal actuators are more bulky,

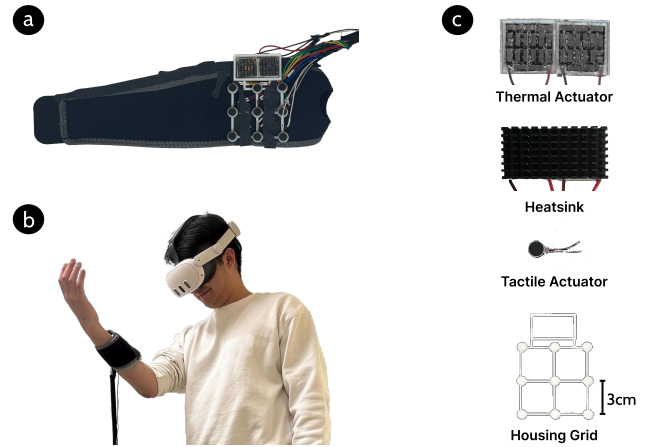
power-hungry, and expensive [7]. Likewise, the slow and spatially diffuse nature of thermal perception complicates spatiotemporal control and multimodal coordination [3, 4]. To address these challenges, several rendering techniques and thermal applications have developed in recent years. Mid-air delivery [11, 12, 18], environmental systems [1, 15], wearable displays [5, 14, 19], and composite illusory effects [8]—thermal feedback has expanded beyond naive contact methods. Notably, the discovery of the thermal masking illusion [13, 17] has enabled precise localization of thermal sensation without perception of the original thermal source, enabling novel effects such as total hand coverage [16] and moving thermal sensations in 1-dimensional lines [10]. However, these works remain limited in expressivity and cannot capture the full context of dynamic thermal sensations. Additionally, these techniques cannot be easily extended to 2D space, as challenges in spatial diffusion, thermal latency, and multimodal coordination result in fragmented and ambiguous percepts. Thus, a need arises in multisensory experiences for compelling methods to render expressive moving thermal sensations, with arbitrary trajectory and dynamic features.

Our demo showcases HeatFlow, a novel thermal-tactile display for rendering dynamic 2D thermal movements (see Figure 1) [9]. By combining the thermal masking [17] and apparent tactile motion [2, 6] illusions, HeatFlow renders salient thermal sensations that move continuously through 2D space with a grid-like array of vibrotactile actuators and a single adjacently-positioned thermal actuator. To drive HeatFlow, we developed a novel control algorithm that determines optimal activation parameters to render arbitrary 2D trajectories. Additionally, we present an thermal flow authoring tool and interactive design experience through HeatFlow. These enable users to create their own unique thermal flows, evaluate their designs with immediate feedback from the HeatFlow display, and directly integrate finalized flows into VR content.

Our comprehensive design pipeline highlights the key benefits of HeatFlow as a multisensory platform for affective haptics and immersive storytelling. The perception-driven HeatFlow approach, enabled by thermal-tactile illusion composition, requires only one thermal actuator for thermal rendering throughout a tactile grid, demonstrating practicality and scalability for system design. The display can also be worn at various body sites for different use cases, indicating broad applicability. The algorithm ensures precise control over sensation movement and compelling realization of complex 2D thermal-tactile coordination with dynamic nuance. Furthermore, design experiences with the authoring tool and VR content promises impactful insights for affective communication and multisensory immersion. Overall, the HeatFlow method represents a practical solution towards expressive thermal feedback in immersive applications, with minimized hardware constraints and rich sensory engagement. For detailed methodology and experimental validation, please refer to [9].

## 2 System

We present an end-to-end flow designer experience to highlight the potential of HeatFlow as a versatile tool for multisensory applications. The customizable wearable display renders compelling thermal flows to various body sites, and the display can be easily integrated with our authoring tool and VR scenarios for custom



**Figure 2: The (a) wearable HeatFlow display, with (b) hardware components and (c) worn by a user.**

flow development. Users will engage with the HeatFlow design suite to create an immersive, multimodal VR experience of their own design.

### 2.1 Hardware

The HeatFlow display consists of a flexible grid of actuators embedded into a wearable sleeve, designed to deliver dynamic thermal-tactile feedback across a 2D cutaneous surface (see Figure 2). The display can be reconfigured to conform to various body sites, including the forearm, palm, and face.

**Customizable Grid** The housing grid is 3D-printed with flexible thermoplastic polyurethane (TPU) plastic, supporting adaptability to different body locations regardless of surface uniformity. The grid structure is designed with 1 mm thickness for lines and 5 mm thickness for actuator housings at points of the grid. This setup ensures all actuators maintain direct skin contact. The original grid follows a  $3 \times 3$  array, but other configurations (eg.  $3 \times 2$ ) can be achieved with the modular design. The actuator grids are securely affixed to off-the-shelf Velcro-based wraps, ensuring comfort in wearability and removability.

**Vibrotactile Actuators** We utilized coin-shaped eccentric rotating mass (ERM) actuators (JIEYI JYC1027) to generate vibrotactile stimulation throughout the actuator grid. These actuators, each 10 mm in diameter and 2.7 mm thick, produce vibrations at 215 Hz with a peak force of 9.8 mN at a voltage of 3 V. The actuators are placed with 3 cm inter-actuator spacing in the primary HeatFlow display, though spacing can decrease to 2.5 cm for smaller body sites. Each actuator is individually controlled with pulse-width modulation (PWM), ensuring precise control over both vibration intensity and duration for nuanced movement patterns. By strategically sequencing actuator activations throughout the 2D grid, the display can induce the perception of rich, moving vibrotactile sensations with dynamic spatiotemporal speed and direction.

**Thermal Actuator** A single Peltier-based thermal actuator (TEGWAY S017A026026) provides thermal sensations in the HeatFlow display. Capable of a maximum temperature differential of

64°C at a voltage of 2.2 V and a current of 6.0 A, the thermal actuator reaches target temperatures in approximately 1 second and cools back to baseline in about 15 seconds. The actuator, fitted with a heatsink, is mounted 1 cm from the adjacent vibrotactile grid, leveraging vibration-induced thermal referral to perceptually shift thermal sensations onto the grid. Combined with tactile motion illusions, this setup enables a single thermal actuator to generate flowing thermal percepts, moving across the 2D surface of the skin.

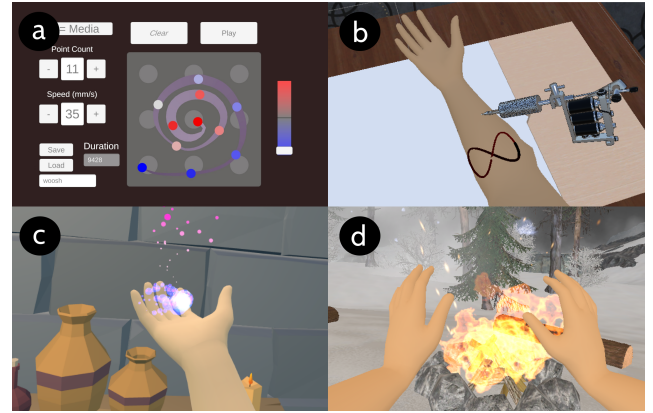
**System Control** The system is controlled by a Raspberry Pi 4 Model B, which precisely orchestrates actuator activations according to motion parameters generated by a host PC and communicated over TCP. The vibrotactile actuators are driven via GPIO with pulse-width modulation (PWM) signals. The polarity and intensity of the thermal actuator is controlled by an H-bridge and programmable power supply (KD6005P, Korad), which communicates with the Raspberry Pi to regulate power input. This system supports real-time control over the HeatFlow display and ensures stable synchronization between tactile motion and thermal stimulation.

## 2.2 Software

HeatFlow employs a novel thermal-tactile control algorithm to produce expressive 2D thermal flows through the display. We provide a design tool to support creation and customization of dynamic thermal flows. Pairing the HeatFlow display and design tool, users can rapidly iterate through various flow patterns and directly integrate their designs with VR content for an immersive, multimodal experience (see Figure 3).

**Algorithm** To produce 2D thermal flows with high perceptual fidelity and salience for, we developed a novel algorithm to generate motion control parameters that perceptually unify tactile motion with thermal sensations through vibration-induced thermal referral. To create a thermal flow, our algorithm (1) selects equidistant phantom points along the input 2D path, (2) calculates target intensities for nearby tactile actuators to create each phantom point, (3) finds instances of intensity conflict throughout the flow, and (4) resolves conflicts with an averaging approach. We addressed limitations in synchronicity of previous work by redefining these actuator polling and interpolation steps, and we enhanced thermal salience by strategically offsetting the activation of the thermal actuator and boosting its input voltage in off-grid locations. With a single thermal actuator and a tactile grid, the HeatFlow algorithm generates perceptual thermal movements that are compelling, dynamic, and expressive.

**Authoring Tool** We present a complementary design tool that enables users to create and test their own custom thermal flows, opening new opportunities for immersive storytelling and affective experiences. Implemented as a tablet-and-pen interface (Samsung Galaxy Tab S9 FE, Samsung S Pen), the tool leverages intuitive drawing-based input methods to support flow design. The design tool includes freehand drawing of flow patterns, controls to adjust motion parameters, per-point modification of thermal and tactile intensity for dynamic sensations, and a direct pipeline to export and synchronize thermal flows to video and virtual reality content. The authoring tool offers a platform for rapid iteration and evaluation



**Figure 3: The (a) design tool interface, which enables creation of custom thermal flows for VR content such as the (b) Tattoo Parlor, (c) Magic Spell, and (d) Winter Camping scenes.**

of thermal flow design, elevating HeatFlow from a display to a multimodal interaction suite.

**VR Scenarios** We provide three unique scenarios for users to experience in VR with their designed thermal flow patterns.

- **Tattoo Parlor** Users see and feel their custom pattern being inked onto their skin as a tattoo. A virtual tattoo machine traces out the pattern with a red line and buzzes while active.
- **Magic Spell** Users conjure a magic spell with their pattern, observing how the magic effect moves and changes over time. The movement path and visual effects of the spell adapts to the parameters of the thermal flow.
- **Winter Camping** Users brave a chilly snow storm at a virtual camping site by stoking a nearby fire. The environmental stimuli of this scene encourages users to explore more abstract and creative thermal patterns.

These VR scenes are designed to support all user-created flows with visual adaptation and seamless integration. Users may pair any of the VR scenes and a body site of their choosing – Forearm, Palm, and Face – with their flow designs, resulting in excitingly custom experiences.

## 3 Demo Experience

At CHI 2026, our demonstration presents an end-to-end design experience to showcase thermal flow creation and rendering through HeatFlow. Attendees will design their own unique thermal flows through the authoring tool, iteratively evaluate their designs with the wearable displays, and directly integrate the final flow into a multimodal VR experience. Attendees will be instructed on how to interact with the HeatFlow system and virtual scenes, including available design parameters and optional audio-visual pairing. Demonstrators will help attendees use the design tools, put on the VR headsets, and attach HeatFlow wearables. The entire demonstration, including setup and briefing, will take approximately 5 minutes per attendee.

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